

Introduction to APIs

1. What is an API? (Simple Explanation)

This playlist is designed to teach:

- What APIs are
- Why they are important
- How to build APIs using **FastAPI (Python framework)**
- How APIs are used in **Machine Learning applications**

So think of this as moving from:

👉 **Basic concept → Real-world usage → Building APIs → ML integration**

2. What is an API?

✅ **Simple Definition:**

API = Application Programming Interface

👉 It is a **bridge** that allows two different software systems to communicate with each other.

💡 **Even Simpler:**

An API lets one program **ask for something** from another program and get a **response**.

📖 **Real Example:**

- You use a weather app
 - The app doesn't store weather data itself
 - It sends a request to a **weather server API**
 - The API returns data (temperature, humidity, etc.)
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🔄 **Basic Flow:**

1. Client (your app/browser) sends a request
 2. API receives it
 3. Server processes it
 4. API sends back a response
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🧠 **Key Idea:**

You don't need to know **how the system works internally**

👉 You just use the API to get results

3. Analogy (Very Important)

Restaurant Example:

- You = Customer
 - Kitchen = System/Server
 - Waiter = API
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Process:

- You tell the waiter your order
 - Waiter takes it to the kitchen
 - Kitchen prepares food
 - Waiter brings food back
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Key Insight:

You **don't go into the kitchen yourself**

 The API (waiter) handles communication

4. Need for APIs

Why do we need APIs?

Without APIs:

- Every system would be isolated
 - No communication between apps
 - Development would be slow and messy
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APIs solve these problems:

1. Enable Communication

Apps can talk to each other

Example:

- Payment system ↔ Bank
 - App ↔ Database
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2. Reusability

You don't rebuild everything

 Use existing APIs (Google Maps, payment gateways, etc.)

3. Faster Development

Instead of building from scratch:

- Just connect APIs
 - Save time and effort
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4. Security & Control

- APIs expose only what's needed
 - Internal system remains hidden
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5. Real-World Examples:

- Login with Google
 - Online payments
 - Weather apps
 - Social media integrations
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5. API from Machine Learning Perspective

This is where things get **very interesting**.

Problem:

Machine Learning models are usually:

- Built in Python
 - Run locally or on servers
 - Not directly usable by apps
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Solution: Use APIs

 Convert your ML model into an API

Example:

You build a **Spam Detection Model**

Without API:

- Only you can use it (inside Python)

With API:

- Any app can use it (website, mobile app, etc.)
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Flow:

1. User enters text
 2. App sends request to ML API
 3. Model processes input
 4. API returns result (Spam / Not Spam)
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Why this matters:

- Makes ML models **usable in real-world applications**
 - Enables:
 - Web apps
 - Mobile apps
 - SaaS products
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FastAPI Introduction

Now comes the tool mentioned in your title.

What is FastAPI?

 **FastAPI** is a modern Python framework used to:

- Build APIs quickly
 - Serve data
 - Deploy ML models
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Why FastAPI is popular:

1. **Very Fast**

- One of the fastest Python frameworks
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2. **Easy to Learn**

- Simple syntax
 - Beginner-friendly
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3. **Automatic Documentation**

- Creates API docs automatically
 - You can test APIs in browser
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4. **Built-in Validation**

- Checks input data automatically
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5. **Perfect for ML**

- Easily connect ML models to APIs
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Simple FastAPI Example:

```
from fastapi import FastAPI

app = FastAPI()
```

```
@app.get("/")
def home():
    return {"message": "Hello World"}
```

What's happening here:

- FastAPI() → creates app
- @app.get("/") → defines endpoint
- Function → returns response

Output:

When you open the URL:
`http://127.0.0.1:8000/`
You get:
`{"message": "Hello World"}`

API Request Types (Extra but Important)

Common methods:

- **GET** → Get data
- **POST** → Send data
- **PUT** → Update data
- **DELETE** → Remove data

Final Understanding (Summary)

👉 API is a **communication bridge** between software systems

👉 It allows:

- Apps to interact
- Data to flow
- Services to connect

👉 In ML:

- APIs make models usable in real applications

👉 FastAPI:

- Helps you **build APIs easily using Python**

One-Line Memory Trick

👉 **API = Request → Process → Response**